Portfolio

By Lindsey Bertin

Overview of Projects

Project 1: Modeling Trust Dynamics in Human-Robot Interaction

Project 2: Designing Social Robots for Mental Health

Project 3: Behavioral and fMRI Analysis of Trust and Reciprocity

Additional Projects and Experience

Project 1: Modeling Trust Dynamics in Human-Robot Interaction

Developing and refining a computational model to simulate how human trust evolves through interactions with robots. The model examines how trust changes when observed robot performance meets, exceeds, or falls short of expectations. Implemented simulations to explore trust dynamics under varying patterns of robot performance and errors.

Background

Current Landscape

- Robots increasingly collaborate with humans in environments requiring effective coordination and appropriate trust
- Human trust is dynamic, shaped by prior expectations and ongoing experiences with robot performance

Knowledge Gap

- Limited understanding of how discrepancies between expected and observed robot performance across various types and severities of errors/successes shape trust over repeated interactions
- Need for computational approaches that capture how trust evolves rather than treating it as a static measure

Future Relevance

- Supports the design of robotic systems that promote appropriately calibrated user trust through reliable and predictable interactions
- Provides a framework for anticipating how different interaction experiences may influence user confidence and reliance
- **Research Question: How do expectations and observed robot performance interact to shape human trust dynamically over repeated human-robot interactions?**

Specs

Role:

- Implemented and refined a computational trust model using simulation-based approaches
- Translated theoretical assumptions about expectations and trust into testable mathematical and computational processes

Methods:

- Developed simulations modeling changes in trust based on discrepancies between expected and observed robot performance
- Manipulated error frequency, severity, and variability to examine simulated trust trajectories across different interaction conditions

Participants:

- Computational modeling study using various previously-published human-robot trust datasets to confirm model accuracy and generalizability

Timeline:

- Ongoing model development, simulation testing, and iterative refinement

Challenge:

- Translating complex psychological constructs such as expectations and trust into measurable computational variables
- Balancing model realism, interpretability, and flexibility across different interaction scenarios

Design & Implementation Takeaways

Experiment Design:

- Developed a computational framework modeling how trust evolves when robot performance meets, exceeds, or violates user expectations
- Built simulations to systematically manipulate performance patterns and examine their influence on modeled trust over repeated interactions

Possible UX Impact:

- Could help identify interaction patterns that support appropriate trust calibration in human–robot systems
- Provides a potential framework for anticipating how system errors and unexpected behaviors influence user confidence
- May inform the design of AI and robotic interfaces that better account for users' expectations and prior experiences

Reflection:

- Strengthened skills in computational modeling, simulation design, iterative testing, and translating behavioral theory into measurable variables
- Reinforced the value of understanding user expectations and modeling behavior over time when evaluating interactive systems

Project 2: Designing Social Robots for Mental Health

Conducted a critical review of animal- and robot-assisted therapy research to identify how interaction design, engagement, and scalability shape therapeutic outcomes. Synthesized clinical findings through a human-factors lens to evaluate which design choices support—or limit—long-term user engagement.

Background

Current Landscape

- Loneliness and social isolation significantly worsen outcomes across multiple mental health conditions, creating demand for scalable interventions
- Animal-assisted therapy (AAT) shows benefits, but faces scalability, cost, and ethical constraints; social robots are increasingly proposed as alternatives

Knowledge Gap

- Clear comparisons between the effectiveness of robot-assisted and animal-assisted therapies across populations
- Understanding of which design and interaction features support sustained user engagement over time.

Future Relevance

- Guide when and how robot-assisted therapy should be deployed in mental healthcare settings
- Inform design priorities for social robots intended for emotionally sensitive use cases

Research Question: How effective are robot-assisted therapies compared to animal-assisted therapies, and what design and human-factors limitations currently constrain their impact?

Specs

Role:

- Primary author, responsible for defining the review scope, framework, and analytical approach
- Synthesized interdisciplinary evidence and translated findings into human-factors and design-relevant insights

Methods:

- Critical narrative literature review of peer-reviewed studies on animal- and robot-assisted therapy
- Comparative analysis across populations, intervention types, and robot design characteristics

Participants:

- Secondary research only; reviewed studies included children, adults, and older adults across psychiatric and neurodevelopmental populations

Timeline:

- Literature review, synthesis, and writing conducted over approximately 12 months

Challenge:

- Highly heterogeneous study designs and outcome measures limited direct comparison
- Many studies prioritized feasibility over long-term effectiveness or engagement

Manuscript available online at https://doi.org/10.1016/j.ajp.2025.104615

Key Insights & Takeaways

Insights:

- Across studies, animal-assisted therapy shows more consistent short-term benefits than robot-assisted therapy, while robot interventions remain promising but uneven in effectiveness and engagement
- The primary gap across robot-assisted therapy research is sustained user engagement, driven by limited adaptability, shallow interaction, and insufficient attention to human factors

UX Impact:

- Identifies limited emotional depth, repetitive interactions, and declining engagement over time as primary UX constraints on the long-term effectiveness of therapeutic robots
- Highlights the need for rigorous UX-focused studies—including controlled trials and longitudinal designs—to evaluate engagement, emotional responsiveness, and real-world use beyond feasibility
- Indicates that effective therapeutic robots must prioritize emotional responsiveness, personalization, and adaptive interaction over anthropomorphic appearance alone

Reflections:

- Emphasized the importance of evaluating user experience outcomes, not just feasibility or novelty, in early-stage human-AI systems
- Reinforced that human factors and interaction design are central to long-term engagement and meaningful impact in emotionally sensitive technologies

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Beyond the human touch: A critical review of the promise and challenges of animal- and robot-assisted therapy in loneliness and mental healthcare

Lindsey Bertin [a], Jeffrey S. Katz [b,c], Gopikrishna Deshpande [b,c], Frank Krueger [a,*]

[a] School of Systems Biology, George Mason University, Fairfax, VA, USA
[b] Department of Psychological Sciences, Auburn University, AL, USA
[c] Auburn University Neuroimaging Center, Department of Electrical & Computer Engineering, Auburn University, AL, USA

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ABSTRACT

Mental health disorders, including depression, anxiety, post-traumatic stress disorder, and dementia, are increasingly recognized as exacerbated by social isolation and loneliness, prompting growing interest in Artificial Intelligence (AI) driven and robotic interventions for social support. Traditional interventions such as animal-assisted therapy (AAT) have demonstrated effectiveness by leveraging the human-animal bond to reduce stress, enhance social engagement, and improve emotional well-being. However, AAT faces logistical challenges, including availability, cost, and animal welfare concerns. In response, robot-assisted therapy (RAT) has emerged as a scalable alternative, offering consistent social interaction and emotional support without the limitations of live animals. Advances in AI have enhanced RAT's potential by enabling more adaptive, engaging interactions, yet questions remain regarding these connections' emotional significance and depth. This review critically evaluates the efficacy of AAT and RAT in addressing mental health disorders, comparing their impact on loneliness, stress, and mental health outcomes. While AAT has demonstrated moderate yet significant benefits across various populations, its accessibility remains limited. Early RAT studies suggest potential in mitigating loneliness and improving mood, particularly in older adults and individuals with neurodevelopmental disorders. However, methodological limitations and uncertainties about emotional depth persist. Future research should enhance robotic interventions through greater emotional responsiveness, personalization, and long-term engagement. Ethical concerns surrounding data privacy and user attachment must be addressed to ensure responsible implementation. By examining the strengths and limitations of both approaches, this review discusses the evolving role of social robotics in mental healthcare, highlighting the need for further investigation to optimize therapeutic potential.

Project 3: Behavioral and fMRI Analysis of Trust and Reciprocity

Analyzed behavioral and fMRI data from a trust game to examine how social decision-making and reciprocity vary across conditions and individuals. Led data preprocessing, statistical analysis, data visualization, and manuscript writing to translate experimental data into interpretable behavioral and neural measures. Research article submitted and currently under review.

Background

Current Landscape

- Trust decisions are shaped by social uncertainty and asymmetric incentives
- Framing effects can shift how people interpret identical outcomes, changing willingness to take social risk

Knowledge Gap

- Few studies have considered the difference between the decision to trust another person versus predicting the trustworthiness of another person
- Less is known about how risk, temptation, and framing effects differ between these two judgements.

Future Relevance

- Clarifying how framing and payoff context shape trust can inform how we design high-stakes choices (e.g., financial, health, platform safety) where users must rely on others
- Identifying behavioral/neural markers of risk-sensitive trust can help predict when trust breaks down—and what conditions support stable cooperation.

Research Question: How do the effects of risk, temptation, and gain vs. loss framing differ between deciding to trust another person and predicting if that person will reciprocate that trust?

Specs

Role:

- Led behavioral and fMRI data analysis, including preprocessing, modeling, and interpretation of trust and reciprocity measures. Developed the full analysis plan and analytical framework for the study
- Wrote manuscript analyzing, visualizing, and interpreting results. Manuscript currently under review.
- Presented findings at NIH BRAIN conference and Society for Neuroscience Annual Meeting

Methods:

- Behavioral analysis of trust decisions and reciprocity predictions across framing and payoff conditions using R
- fMRI preprocessing and analysis using SPM12, MATLAB, and Python with planned group-level analyses linking neural activity to decision variables

Participants:

- Existing dataset collected in a prior fMRI trust game study; undergraduate students scanned across two task runs with multiple trial types

Timeline:

- Research, manuscript writing, and journal submission conducted over approximately 12 months

Challenge:

- Designing an analysis pipeline that integrates behavioral and neural data while maintaining interpretability
- Managing analytic complexity across multiple conditions, runs, and outcome measures

Code and datasets are available here: https://doi.org/10.5281/zenodo.23987481 (published anonymously as manuscript is still under review)

Analysis Focus & Early Takeaways

Analysis Focus:

- Quantifying how gain vs. loss framing and payoff structure shape trust decisions and expectations of reciprocity
- Linking trial-level decision and prediction variables to neural responses associated with risk evaluation and social decision-making

Future UX Impact:

- Shows that framing and payoff structure meaningfully shape trust decisions and expectations of reciprocity, even when outcomes are identical
- Highlights trust as a dynamic, risk-sensitive process rather than a stable user trait, with implications for system design under uncertainty
- Informs the design of high-stakes systems (e.g., financial, health, safety) where framing can unintentionally amplify perceived social risk

Reflection:

- Reinforced the value of integrating behavioral and neural data to interpret trust-related decisions more precisely
- Strengthened skills in designing interpretable analysis pipelines for complex, multi-condition datasets

Dissociation Between Trust Decisions and Social Inference

Lindsey Bertin[1], Frank Krueger[1,2]
[1]Interdisciplinary Program in Neuroscience, George Mason University, USA, lbertin@gmu.edu, [2]School of Systems Biology, George Mason University, USA

Abstract

Trust requires accepting vulnerability to another person's choices. This study examined how risk, temptation, and gain/loss framing influence trust decisions and reciprocity predictions. Eighty-one participants completed a binary trust game during fMRI, deciding whether to trust a partner before predicting whether the partner would reciprocate. Behavioral analyses showed that trust increased under low risk and low temptation, whereas reciprocity expectations increased under low temptation. Framing influenced trust primarily under high-risk conditions. Risk had a greater influence on trust decisions, whereas temptation more strongly influenced reciprocity predictions, suggesting distinct processes underlying trust and social inference. Neurally, low temptation during trust decisions was associated with mPFC activity, while reciprocity prediction engaged posterior parietal cortex, consistent with integration of risk and partner temptation. Together, these findings suggest that trust decisions and reciprocity predictions rely on partially distinct behavioral and neural mechanisms.

Methods

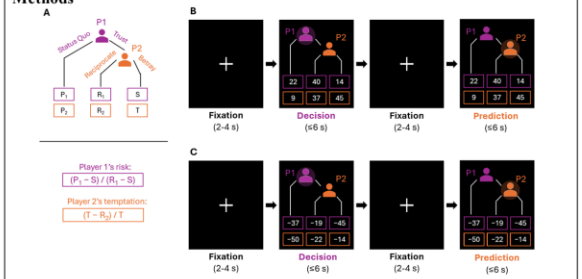


Fig. 1. Experimental paradigm. (A) *Trust Game and Payoff Structure* P1 chose the status quo or trusted P2; after trust, the outcome depended on whether P2 reciprocated or betrayed. Risk and temptation were calculated from the original gain-frame payoff matrices for both framing conditions. Loss-frame matrices were generated by subtracting S + T from each gain-frame payoff, preserving the underlying Risk and Temptation values. **(B)** *Gain-frame trial*. Each trial consisted of a fixation, the Decision stage, a second fixation, and the Prediction stage. **(C)** *Loss-frame trial*. Identical structure; participants began each session with 9,500 points, and outcomes were presented as losses.

Behavioral Results: Reciprocity Prediction

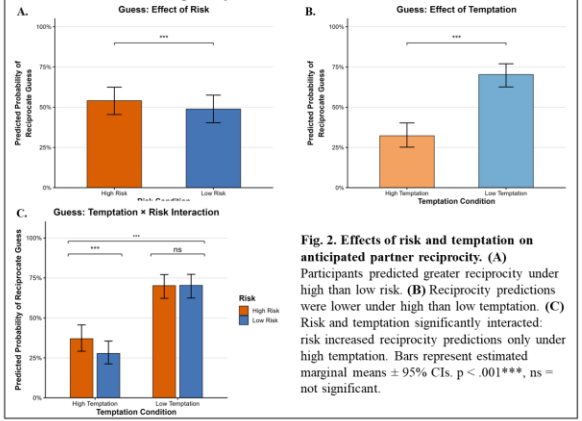


Fig. 2. Effects of risk and temptation on anticipated partner reciprocity. (A) Participants predicted greater reciprocity under high than low risk. **(B)** Reciprocity predictions were lower under high than low temptation. **(C)** Risk and temptation significantly interacted: risk increased reciprocity predictions only under high temptation. Bars represent estimated marginal means ± 95% CIs. $p < .001$***, ns = not significant.

Behavioral Results: Trust Decision

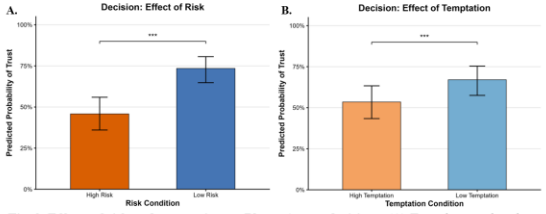


Fig. 3. Effects of risk and temptation on Player 1 trust decisions. (A) Trust decreased under high compared with low risk. **(B)** Trust decreased under high compared with low temptation. Bars represent estimated marginal means ± 95% CIs. $p < .001$***.

Behavioral Results: Cross-Stage Dissociation

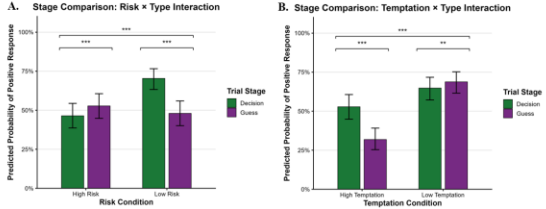


Fig. 4. Comparison of trust decisions and reciprocity predictions across risk and temptation conditions. (A) Risk influenced trust decisions and reciprocity predictions differently: trust was higher under low than high risk, whereas reciprocity predictions were higher under high than low risk. **(B)** Temptation reduced both trust and reciprocity predictions, with a stronger effect on trust decisions. Bars represent estimated marginal means ± 95% CIs. $p < .01$**, $p < .001$***.

Conclusions

- Framing influenced trust primarily under high-risk conditions.
- Low temptation increased trust, reciprocity expectations, and mPFC activity.
- **Trust decisions and reciprocity prediction rely on distinct behavioral and neural mechanisms: risk primarily guides trust decisions, whereas temptation drives social inference.**

Neuroimaging Results: Trust Decision & Reciprocity Prediction

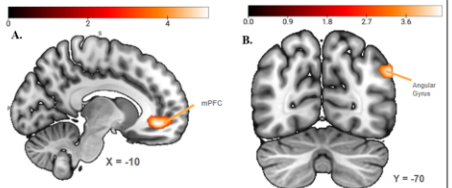


Fig. 5. Neural correlates of temptation during trust decisions and reciprocity prediction. (A) Low temptation elicited greater mPFC activation during trust decisions. **(B)** A Risk × Temptation interaction in the left angular gyrus indicated that neural responses to temptation depended on risk during reciprocity prediction. All clusters survived FWE correction ($p < .05$). Color bars represent increasing T-values.

Neuroimaging Results: Cross-Stage Dissociation

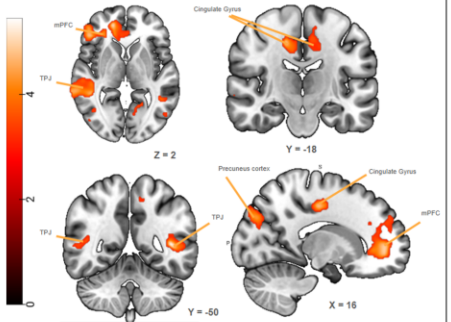


Fig. 6. Stage-dependent neural processing of temptation. Greater temptation-related activation during reciprocity prediction than trust decisions was observed in the bilateral TPJ, mPFC, precuneus, and cingulate cortex. All clusters survived FWE correction ($p < .05$). Color scale indicates T-values.

References

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Evans, A. M., & van Beest, I. (2017). Gain-loss framing effects in dilemmas of trust and reciprocity. Journal of Experimental Social Psychology, 73, 151–163. https://doi.org/10.1016/j.jesp.2017.06.012
Fang, H. (2024). Neurocomputational Mechanism of Reciprocity: The Core and Periphery (Doctoral dissertation, University of Mannheim).

Additional Projects and Experience

Designed Study on the Effect of AI Errors on User Compliance and Reliance

- Designed and implemented a web-based human–AI interaction experiment to examine how different AI error types influence user trust, reliance, compliance, and correction behavior
- Built the full study in jsPsych and JATOS, including task flow, AI feedback logic, knowledge checks, and post-task surveys.

Participant Recruitment & Study Support

- Developed AI agent to assist with research pipelines
- Supported participant recruitment and coordination for an ongoing behavioral research study, ensuring protocol adherence and clear participant communication
- Recruited, interviewed, trained, and managed a team of undergraduate research assistants.

Diabetes Educational Program Evaluation Volunteer Researcher

- Analysis of survey and medical data from diabetes educational program attendees. Coded responses from qualitative data, analyzed quantitative clinical metrics, used statistical analysis and data visualization to demonstrate trends across fiscal years
- Generated presentations with research findings to evaluate program efficacy and recommended changes to increase engagement, resource adoption, health outcomes, and perceived helpfulness among attendees.